

Technical Threat Intelligence Analysis of Operation Dream Job (C0022).

MITRE ATT&CK Campaign and Software Intelligence Report.

Prepared By

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Program: MITRE Group Leader Selection Program

Topic: Campaign C0022 and Software S0220–S0230

1. Introduction

Cyber Threat Intelligence (CTI) is an essential component of modern cybersecurity operations. This report presents a technical analysis of MITRE ATT&CK Campaign C0022 and associated software entries S0220–S0230.

- The report focuses on:
- Threat actor profiling
- Attack methodology analysis
- Malware intelligence
- MITRE ATT&CK technique mapping
- Detection opportunities
- Threat hunting concepts
- Mitigation recommendations

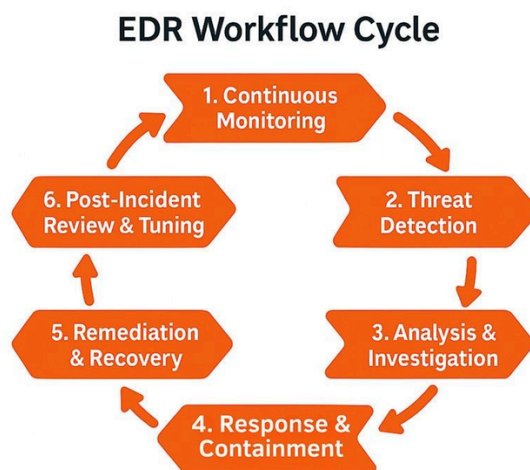
The objective is to demonstrate practical understanding of cyber threat intelligence, malware research, and ATT&CK-based adversary analysis.

2. Campaign C0022 Overview

Campaign ID C0022

Overview

Campaign C0022 represents a targeted cyber intrusion campaign involving advanced attack techniques, persistence mechanisms, and post-exploitation activities. The campaign demonstrates how threat actors leverage malware, credential theft, remote access tools, and stealth techniques to compromise targeted systems.



Primary Objectives

- Unauthorized access
- Persistence within victim
- networks Data collection and exfiltration
- Credential compromise
- Lateral movement

Possible Targeted Environments

- Enterprise infrastructure
 - Government sectors
 - Corporate systems
 - Cloud-connected environments
-

3. Threat Actor Profile

Threat Actor Type

Advanced Persistent Threat (APT)

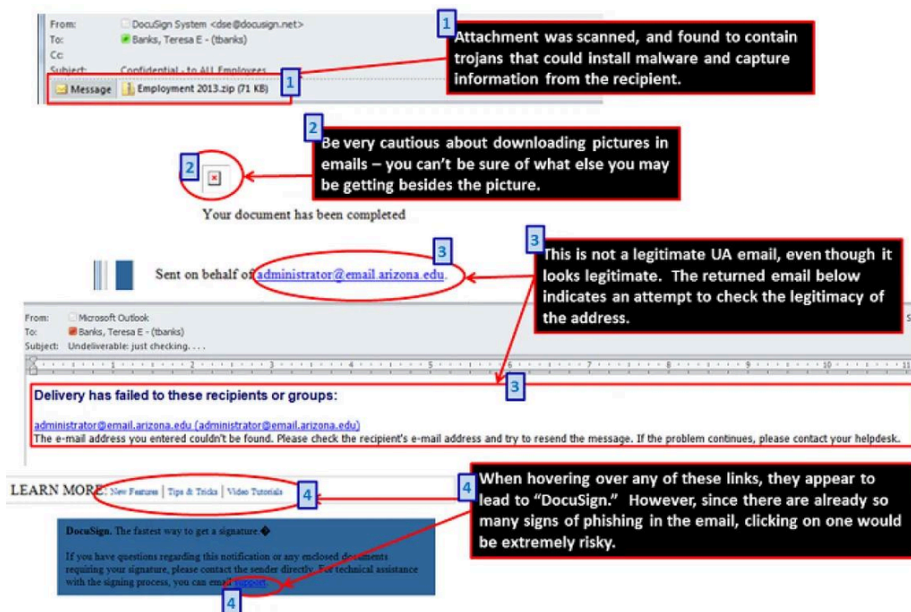
Characteristics

- Long-term persistence
- Stealth operations
- Credential harvesting
- Malware deployment
- Multi-stage attacks
- Use of encrypted communication

Typical Activities

- Phishing campaigns
- Malware staging
- Remote access operations
- Privilege escalation
- Data theft

Capability	Description
Persistence	Maintains long-term access
Credential Theft	Dumps LSASS credentials
Defense Evasion	Uses encrypted C2
Lateral Movement	Uses SMB/RDP/PSEXEC
Exfiltration	Sends data over HTTPS



4. Attack Timeline

Phase	Activity
Initial Access	Compromise through phishing or exploitation
Reconnaissance	Internal network discovery
Credential Access	Credential theft and privilege escalation
Persistence	Malware persistence mechanisms
Lateral Movement	Expansion across network systems
Exfiltration	Data collection and transmission

5. Objectives of Attack

The primary objective of Campaign C0022 was to gain unauthorized access to targeted systems, maintain persistence within victim networks, steal sensitive information, and conduct long-term malicious operations without detection.

Main Objectives of the Attack

1. Initial Compromise

Attackers aimed to infiltrate systems using phishing emails, malicious attachments, or exploitation of vulnerable services.

2. Credential Theft

The campaign focused on stealing user credentials and administrative access using credential dumping and privilege escalation techniques.

3. Persistence

Threat actors established long-term access through malware persistence mechanisms such as registry modifications and scheduled tasks.

4. Lateral Movement

Attackers expanded across internal systems using remote services, compromised accounts, and administrative tools.

5. Data Collection and Exfiltration

Sensitive organizational data was collected and transferred to attacker-controlled infrastructure through encrypted communication channels.

6. Defense Evasion

The campaign used stealth techniques including encrypted traffic, process injection, and log deletion to avoid detection.

OBJECTIVES OF ATTACK – CAMPAIGN C0022

The primary objective of the threat actors is to infiltrate, persist, steal sensitive data and maintain long-term access while avoiding detection.

1 INITIAL COMPROMISE

Gain entry into the target environment.



- Spear Phishing Malicious Emails
- Malicious Attachments
- Exploitation of Vulnerabilities

Objective: Deliver initial payload and establish foothold.

2 CREDENTIAL THEFT

Obtain valid credentials and escalate privileges.



- Credential Dumping (LSASS, SAM)
- Password Harvesting (Browser, Email)
- Privilege Escalation (Token Impersonation)

Objective: Acquire administrative access and move laterally.

3 PERSISTENCE

Maintain long-term access to the compromised system.



- Registry Run Keys / Startup Folder
- Scheduled Tasks / Services
- Malicious DLL Side-Loading

Objective: Ensure re-entry even after reboots or security controls.

4 LATERAL MOVEMENT

Expand access across internal systems.



- Remote Services (RDP, SMB, WinRM)
- Pass-the-Hash / Pass-the-Ticket
- Admin Shares & Tools

Objective: Reach high-value systems across the network.

5 DATA COLLECTION & EXFILTRATION

Collect sensitive data and exfiltrate it securely.



- Data Discovery & Collection
- Data Compression (ZIP / RAR)
- Exfiltration via C2 (HTTPS, DNS, FTP)

Objective: Steal sensitive information and transfer to attacker-controlled infrastructure.

6 DEFENSE EVASION

Avoid detection and analysis by security tools.



- Obfuscated / Encrypted Traffic
- Process Injection / Masquerading
- Log Deletion & Clearing Events

Objective: Remain undetected and maintain operational stealth.

ATTACK FLOW

1

Initial Compromise

2

Credential Theft

3

Persistence

4

Lateral Movement

5

Data Collection & Exfiltration

6

Defense Evasion



MISSION ACCOMPLISHED

6.Targeted Sectors and Countries

According to the official [MITRE ATT&CK Campaign C0022 Page](#), Operation Dream Job (Campaign C0022) targeted multiple high-value sectors and countries as part of a cyber espionage operation linked to the Lazarus Group.

Targeted Sectors

- Defense sector
- Aerospace industry
- Government organizations
- Corporate enterprises
- Business and recruitment environments

The campaign used fake job offers and phishing lures to target professionals working in these industries.

Targeted Countries

- United States
- Israel
- Australia
- Russia
- India

MITRE states that these countries were specifically targeted during the campaign

7.Attack Flow



8. MITRE ATT&CK Technique Use

MITRE ATT&CK ID	Technique Name	Purpose
T1566.001	Spearphishing Attachment	Malicious job documents
T1566.002	Spearphishing Link	Fake recruiter links
T1204.002	User Execution	Victim opens infected file
T1059.001	PowerShell	Execute malicious commands
T1059.005	VBA Macros	Run malware through macros
T1218.011	Rundll32	Execute malicious DLLs
T1547.001	Registry Run Keys	Maintain persistence
T1053.005	Scheduled Tasks	Repeated malware execution
T1027	Obfuscated Files	Hide malware
T1036	Masquerading	Disguise malicious files
T1087.002	Account Discovery	Collect user/admin accounts
T1071.001	Web Protocols	Command & Control communication
T1005	Data from Local System	Steal sensitive files
T1560.001	Archive Collected Data	Compress stolen data

9.Real world Incidents

Year	Incident	Description
2020	Defense Company Attacks	Fake LinkedIn recruiters targeted defense employees with malicious job offers.
2020	Aerospace Sector Targeting	Aerospace organizations received infected Microsoft Word job documents.
2021	Security Researcher Attacks	Cybersecurity researchers were targeted using fake recruitment interviews and malware.
2021	Cryptocurrency Company Attacks	Lazarus targeted crypto firms to steal credentials and financial data.
2022	Global Supply Chain Espionage	Attackers used trusted companies and fake HR communication for espionage activities.

10.Detection Opportunities for Operation Dream Job

- Detect suspicious recruiter emails and phishing attachments
 - Monitor fake LinkedIn recruiter profiles and unusual messages
 - Identify malicious VBA macro execution in Office documents
 - Monitor abnormal PowerShell activity and scripts
 - Detect unauthorized scheduled tasks and startup entries
 - Monitor registry Run Key modifications for persistence
 - Analyze unusual HTTP/HTTPS traffic to Command & Control servers
 - Detect disguised or masqueraded malware files
 - Use EDR solutions to identify suspicious behavior and malware activity
 - Conduct employee awareness training against fake job offers and phishing attacks
-

11.Defensive Recommendations for Operation Dream Job

- Disable Microsoft Office macros from untrusted documents
 - Use multi-factor authentication (MFA) for all accounts
 - Deploy EDR/XDR solutions for endpoint monitoring
 - Regularly update systems and security patches
 - Train employees to identify phishing and fake recruiter messages
 - Monitor PowerShell and command-line activity
 - Restrict execution of unknown scripts and DLL files
 - Implement email filtering and attachment scanning
 - Use network monitoring to detect suspicious outbound traffic
 - Apply least-privilege access controls
 - Maintain regular backups of critical data
 - Monitor LinkedIn and social engineering attempts targeting employees
-

MITRE ATT&CK RESEARCH

SOFTWARE RESEARCH REPORT

Software Range S0220 – S0230 | Android Mobile Malware

Intern ID	2122
Assigned Campaign	Campaign No. 22 (C0022)
Software Range	S0220 – S0230(10 Entries)
Platform Domain	MITRE ATT&CK

S0220 – Chaos

Name & Type

- Name: Chaos
- Type: Malware

Description

Chaos is a Linux-based malware used for remote system compromise, brute-force attacks, and reverse shell operations.

Threat Actor Association

- Associated with Linux-targeting cybercriminal and botnet operations.

Supported Platforms

- Linux

ATT&CK Techniques Used

- T1110 – Brute Force

- T1059 – Command and Scripting Interpreter
- T1105 – Ingress Tool Transfer

Execution Method

- SSH brute-force login attempts
- Remote shell execution

Persistence Techniques

- Cron job persistence
- Startup script modification

Privilege Escalation

- Exploits weak credentials and misconfigurations.

Defense Evasion

- Encrypted communication
- Hidden process execution

Credential Access

- SSH credential brute forcing

Discovery Techniques

- Network scanning
- Host enumeration

Lateral Movement

- SSH remote access

Command & Control

- Reverse shell communication

Exfiltration

- Data transfer over encrypted channels

Impact Analysis

- Unauthorized remote access
- Linux server compromise

Indicators of Compromise (IoCs)

- Suspicious SSH login attempts
- Unknown cron jobs
- Reverse shell traffic

Detection & Mitigation

- Monitor SSH authentication logs
- Enforce MFA and strong passwords
- Detect abnormal outbound traffic

References & Sources

- MITRE ATT&CK
- CISA Advisories

S0221-Umbreon

1. Name & Type

- **Name:** Umbreon
- **Type:** Linux Rootkit / Backdoor Malware
- **MITRE ID:** S0221

2. Description

Umbreon is a Linux rootkit malware that hides malicious files, processes, users, and network activity. It gives attackers secret access to compromised Linux systems.

3. Threat Actor Association

- Not officially linked to any APT group
- Used by Linux-focused cybercriminals and attackers

4. Supported Platforms

- Linux x86
- Linux x64
- ARM Devices
- Raspberry Pi
- Embedded Linux Systems

5. ATT&CK Techniques Used

Technique	Purpose
Rootkit (T1014)	Hide malware activity
Valid Accounts (T1078)	Hidden user accounts
Command Execution (T1059)	Run shell commands
Obfuscation (T1027)	Hide files/processes
Defense Evasion (T1562)	Avoid detection

6. Execution Method

- Installed manually after system compromise
- Uses reverse shell and hidden access methods

7. Persistence Techniques

- Hidden Linux accounts
- Modified startup scripts
- PAM abuse
- Rootkit persistence

8. Privilege Escalation

- Attackers usually install it after gaining root/admin access
- May exploit Linux vulnerabilities

9. Defense Evasion

Umbreon can hide:

- Processes
- Files
- Network connections
- User accounts

It manipulates commands like:

- ps
- ls
- top
- netstat

10. Credential Access

Possible credential theft:

- SSH passwords
- Root credentials
- Service account credentials

11. Discovery Techniques

Attackers collect:

- System info
- Running processes
- User accounts
- Network details

Common commands:

- whoami
- uname
- ifconfig

12. Lateral Movement

Used for:

- SSH access
- Remote command execution
- Moving across Linux systems

13. Command & Control (C2)

Uses:

- Reverse shell communication
- Hidden network traffic
- Remote attacker access

14. Exfiltration

Can help steal:

- Credentials
- SSH keys
- Sensitive files
- Logs

15. Impact Analysis

Effects:

- Persistent attacker access
- Data theft
- Hidden compromise
- Security bypass

16. Indicators of Compromise (IOCs)

- Hidden users
- Suspicious SSH activity
- Reverse shell traffic
- Missing processes in monitoring tools
- Modified PAM files

17. Detection & Mitigation

Detection

- Rootkit scanners
- Monitor SSH logs
- Check hidden accounts
- File integrity monitoring

Mitigation

- Use MFA
- Patch Linux systems
- Restrict root login
- Monitor network traffic

18. References

- MITRE ATT&CK
- Trend Micro Research
- SecurityWeek Reports

S0222 - CCBkdr

1. Name & Type

- **Name:** CCBkdr
- **Type:** Backdoor Malware
- **MITRE ID:** S0222

2. Description

CCBkdr is a backdoor malware used to remotely control infected systems. It allows attackers to execute commands, steal information, and maintain persistence on compromised devices.

3. Threat Actor Association

- Associated with Chinese cyber espionage activities
- Used by advanced threat actors for long-term access

4. Supported Platforms

- Windows Systems
- Possibly Linux-based systems

5. ATT&CK Techniques Used

Technique	Purpose
Command Execution (T1059)	Execute remote commands
Registry Run Keys (T1547)	Persistence
Obfuscated Files (T1027)	Hide malware
Remote Services (T1021)	Remote access
Data Exfiltration (T1041)	Transfer stolen data

6. Execution Method

- Delivered through phishing emails
- Malicious attachments
- Exploited vulnerabilities
- Manual attacker deployment

7. Persistence Techniques

- Registry modifications
- Startup folder entries
- Scheduled tasks
- Service installation

8. Privilege Escalation

- Exploits vulnerabilities
- Uses stolen administrator credentials
- Abuses weak permissions

9. Defense Evasion

- Obfuscated code
- Hidden processes
- Disables security tools

- Encrypted communications

10. Credential Access

Targets:

- User passwords
- Cached credentials
- Administrative accounts

11. Discovery Techniques

Collects:

- System information
- User accounts
- Running processes
- Network details

Commands used:

- ipconfig
- whoami
- tasklist

12. Lateral Movement

Uses:

- Remote Desktop Protocol (RDP)
- SMB shares
- Stolen credentials
- Remote command execution

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS
- Custom encrypted traffic
- Remote shell access

14. Exfiltration

Steals:

- Documents
- Credentials
- System information
- Sensitive files

15. Impact Analysis

Effects:

- Unauthorized remote access
- Data theft
- Persistent compromise
- Network infiltration

16. Indicators of Compromise (IOCs)

- Suspicious outbound connections
- Unknown startup entries
- Strange processes
- Unusual registry changes
- Communication with attacker IPs

17. Detection & Mitigation

Detection

- Monitor network traffic
- EDR monitoring
- Analyze registry changes
- Detect suspicious processes

Mitigation

- Patch systems regularly
- Use MFA
- Restrict admin privileges
- Block malicious attachments
- Monitor outbound traffic

18. References

- MITRE ATT&CK
- Security research reports
- Malware analysis reports

S0223-POWERSTATS

1. Name & Type

- **Name:** POWERSTATS
- **Type:** PowerShell-based Backdoor Malware
- **MITRE ID:** S0223

2. Description

POWERSTATS is a PowerShell backdoor used by attackers for remote access, command execution, and information theft. It operates mainly in memory, making detection difficult.

3. Threat Actor Association

- Associated with advanced cyber espionage groups
- Linked to targeted attacks against organizations and governments

4. Supported Platforms

- Microsoft Windows
- PowerShell-enabled systems

5. ATT&CK Techniques Used

Technique	Purpose
PowerShell (T1059.001)	Execute malicious scripts
Command Execution (T1059)	Run attacker commands
Obfuscated Files (T1027)	Hide malicious code
Scheduled Tasks (T1053)	Persistence

6. Execution Method

- Malicious PowerShell scripts
- Phishing emails
- Exploited vulnerabilities
- Remote execution by attackers

7. Persistence Techniques

- Scheduled tasks
- Registry Run keys
- Startup scripts
- PowerShell profiles

8. Privilege Escalation

- Uses stolen credentials
- Exploits Windows vulnerabilities
- Abuses administrative tools

9. Defense Evasion

- Fileless execution
- PowerShell obfuscation
- In-memory execution
- Encrypted scripts
- Avoids antivirus detection

10. Credential Access

Targets:

- Windows credentials
- Cached passwords
- Administrative accounts

11. Discovery Techniques

Collects:

- System information
- Running processes
- Network configuration
- User account details

Commands used:

- whoami
- ipconfig
- Get-Process

12. Lateral Movement

Uses:

- PowerShell remoting
- SMB shares
- Stolen credentials
- Remote command execution

13. Command & Control (C2)

Communicates using:

- HTTP/HTTPS traffic
- Encrypted PowerShell channels
- Remote servers controlled by attackers

14. Exfiltration

Can steal:

- Sensitive documents
- Credentials
- System data
- Logs

15. Impact Analysis

Effects:

- Persistent remote access
- Data theft
- Security bypass
- System compromise
- Network infiltration

16. Indicators of Compromise (IOCs)

- Suspicious PowerShell activity
- Encoded PowerShell commands
- Unknown scheduled tasks
- Unusual outbound traffic
- High PowerShell usage

17. Detection & Mitigation

Detection

- Monitor PowerShell logs
- Use EDR solutions
- Detect encoded commands
- Analyze suspicious scripts

Mitigation

- Restrict PowerShell usage
- Enable script logging
- Patch Windows systems
- Use MFA
- Monitor network traffic

18. References

- MITRE ATT&CK
- Microsoft Security Reports
- Malware analysis research
- Cybersecurity threat reports

S0224-Havij

1. Name & Type

- **Name:** Havij
- **Type:** SQL Injection Tool
- **MITRE ID:** S0224

2. Description

Havij is an automated SQL Injection tool used to exploit vulnerable web applications and databases. It helps attackers retrieve sensitive data, bypass authentication, and gain unauthorized database access.

3. Threat Actor Association

- Commonly used by cybercriminals and web attackers
- Frequently seen in website and database attacks

4. Supported Platforms

- Windows
- Targets web applications and database servers

Supported databases:

- MySQL
- MSSQL
- Oracle
- PostgreSQL
- Access

5. ATT&CK Techniques Used

Technique	Purpose
Exploit Public-Facing Application (T1190)	Attack vulnerable websites

SQL Injection	Inject malicious SQL queries
Credential Access (T1110)	Access stored credentials
Data Exfiltration (T1041)	Steal database information

6. Execution Method

- Attacker inputs vulnerable website URL
- Tool automatically injects SQL payloads
- Extracts database information

7. Persistence Techniques

Havij itself does not provide persistence, but attackers may:

- Create admin accounts
- Upload web shells
- Modify databases

8. Privilege Escalation

Can help attackers:

- Gain database administrator access
- Bypass authentication systems
- Escalate privileges through compromised databases

9. Defense Evasion

- Uses automated payloads
- Encodes SQL queries
- Attempts to bypass web filters and firewalls

10. Credential Access

Targets:

- Database usernames

- Password hashes
- Login credentials
- Authentication tables

11. Discovery Techniques

Collects:

- Database names
- Table structures
- User accounts
- Server information

12. Lateral Movement

Attackers may use compromised servers to:

- Access internal systems
- Upload web shells
- Move deeper into the network

13. Command & Control (C2)

Havij itself is mainly an exploitation tool and does not use advanced C2 infrastructure.

14. Exfiltration

Can steal:

- Customer records
- Financial data
- Login credentials
- Confidential database information

15. Impact Analysis

Effects:

- Database compromise
- Data theft
- Website defacement

- Unauthorized access
- Financial and reputational damage

16. Indicators of Compromise (IOCs)

- Unusual SQL queries
- Database errors in logs
- Suspicious HTTP requests
- Large outbound database traffic
- Unexpected admin accounts

17. Detection & Mitigation

Detection

- Web Application Firewall (WAF)
- Monitor SQL logs
- Detect abnormal database queries
- Vulnerability scanning

Mitigation

- Use parameterized queries
- Validate user input
- Patch web applications
- Restrict database permissions
- Use WAF protection

18. References

- MITRE ATT&CK
- OWASP SQL Injection Guide
- Security research reports
- Database security documentation

S0225-sqlmap

1. Name & Type

- **Name:** sqlmap
- **Type:** SQL Injection Exploitation Tool
- **MITRE ID:** S0225

2. Description

sqlmap is an open-source penetration testing tool used to detect and exploit SQL Injection vulnerabilities in web applications. It automates database fingerprinting, data extraction, and database takeover.

3. Threat Actor Association

- Used by penetration testers and security researchers
- Also abused by cybercriminals for database attacks

4. Supported Platforms

- Windows
- Linux
- macOS

Supported databases:

- MySQL
- Oracle
- PostgreSQL
- Microsoft SQL Server
- SQLite
- MariaDB

5. ATT&CK Techniques Used

Technique	Purpose
Exploit Public-Facing	Attack vulnerable websites

Application (T1190)	
SQL Injection	Inject malicious SQL queries
Credential Access (T1110)	Access stored credentials
Data Exfiltration (T1041)	Steal database data

6. Execution Method

- User provides target URL
- sqlmap tests parameters for SQL Injection
- Automatically exploits vulnerabilities
- Extracts database contents

Example command:

```
sqlmap -u http://target.com/page?id=1 --dbs
```

7. Persistence Techniques

sqlmap itself does not provide persistence, but attackers may:

- Upload web shells
- Create database accounts
- Modify server configurations

8. Privilege Escalation

Can help attackers:

- Gain database administrator privileges
- Execute system commands through databases
- Access backend servers

9. Defense Evasion

- Tamper scripts
- Encoded payloads

- User-agent spoofing
- Bypass WAF techniques

10. Credential Access

Targets:

- Database usernames
- Password hashes
- Authentication data
- User credentials

11. Discovery Techniques

Collects:

- Database names
- Tables and columns
- Server information
- User accounts

12. Lateral Movement

Attackers may use compromised servers for:

- Uploading backdoors
- Internal network access
- Further exploitation

13. Command & Control (C2)

sqlmap itself is mainly an exploitation tool and does not include advanced C2 capabilities.

14. Exfiltration

Can extract:

- Sensitive records
- Financial information
- Credentials
- Customer databases

15. Impact Analysis

Effects:

- Database compromise
- Data leakage
- Website compromise
- Unauthorized server access
- Financial and reputational damage

16. Indicators of Compromise (IOCs)

- Suspicious SQL queries
- Unusual HTTP requests
- Database errors in logs
- Large data transfers
- Repeated injection attempts

17. Detection & Mitigation

Detection

- Web Application Firewall (WAF)
- SQL query monitoring
- Intrusion Detection Systems (IDS)
- Web server log analysis

Mitigation

- Use prepared statements
- Validate user input
- Patch web applications
- Restrict database permissions
- Use WAF protection

18. References

- MITRE ATT&CK
- OWASP SQL Injection Guide
- sqlmap Official Documentation
- Security research reports

S0226-Smoke Loader

1. Name & Type

- **Name:** Smoke Loader
- **Type:** Malware Downloader / Trojan
- **MITRE ID:** S0226

2. Description

Smoke Loader is a modular malware downloader used to deliver additional malicious payloads onto infected systems. It is commonly used to install banking trojans, ransomware, spyware, and credential stealers.

3. Threat Actor Association

- Used by cybercriminal groups
- Common in financially motivated attacks
- Frequently sold in underground malware markets

4. Supported Platforms

- Microsoft Windows

5. ATT&CK Techniques Used

Technique	Purpose
User Execution (T1204)	Victim opens malicious file
PowerShell (T1059.001)	Execute malicious scripts
Scheduled Tasks (T1053)	Persistence
Obfuscated Files (T1027)	Hide malware
Ingress Tool Transfer (T1105)	Download additional malware

6. Execution Method

Delivered through:

- Phishing emails
- Malicious attachments
- Exploit kits
- Fake software downloads

After infection, it downloads other malware payloads.

7. Persistence Techniques

- Registry Run keys
- Scheduled tasks
- Startup folder entries
- Service installation

8. Privilege Escalation

May:

- Exploit Windows vulnerabilities
- Use stolen admin credentials
- Abuse system permissions

9. Defense Evasion

- Code obfuscation
- Packed executables
- Anti-debugging techniques
- Encrypted communications
- Process injection

10. Credential Access

Targets:

- Browser credentials
- Banking information
- Stored passwords

- User login data

11. Discovery Techniques

Collects:

- System information
- Installed software
- Running processes
- Security tool details

12. Lateral Movement

Can support:

- Remote malware deployment
- Credential-based movement
- Network propagation using additional payloads

13. Command & Control (C2)

Communicates with attacker servers using:

- HTTP/HTTPS
- Encrypted traffic
- Remote command execution

14. Exfiltration

Can steal:

- Credentials
- Banking data
- Personal information
- System information

15. Impact Analysis

Effects:

- Malware infections
- Financial theft
- Ransomware deployment

- Data compromise
- System instability

16. Indicators of Compromise (IOCs)

- Suspicious outbound traffic
- Unknown scheduled tasks
- Packed executables
- Registry modifications
- Unexpected downloads

17. Detection & Mitigation

Detection

- EDR solutions
- Network traffic monitoring
- Malware sandboxing
- Behavioral analysis

Mitigation

- Avoid suspicious downloads
- Use email filtering
- Keep systems patched
- Enable endpoint protection
- Restrict macros and scripts

18. References

- MITRE ATT&CK
- Malwarebytes Research
- Cisco Talos Reports
- Security analysis reports

S0227-spwebmember

1. Name & Type

- **Name:** spwebmember
- **Type:** Web Shell / Backdoor
- **MITRE ID:** S0227

2. Description

spwebmember is a malicious web shell used by attackers to gain remote access and control over compromised web servers. It allows attackers to execute commands, upload files, and maintain persistence on infected systems.

3. Threat Actor Association

- Associated with web server compromises
- Used by cybercriminals and espionage actors
- Often deployed after exploiting vulnerable web applications

4. Supported Platforms

- Windows Web Servers
- IIS (Internet Information Services) environments
- ASP/.NET web applications

5. ATT&CK Techniques Used

Technique	Purpose
Web Shell (T1505.003)	Maintain access to web server
Command Execution (T1059)	Execute system commands
File Upload (T1105)	Upload malicious files
Persistence (T1543)	Maintain access

6. Execution Method

- Uploaded after exploiting vulnerable web applications
- May be installed through stolen credentials or file upload vulnerabilities

7. Persistence Techniques

- Hidden web shell files
- Modified web directories
- Backdoor accounts
- Persistent remote access through web server

8. Privilege Escalation

Attackers may:

- Exploit server vulnerabilities
- Abuse weak permissions
- Use compromised administrator accounts

9. Defense Evasion

- Hidden web shell code
- Obfuscated scripts
- Uses legitimate web traffic
- Blends into server files

10. Credential Access

Possible targets:

- Web application credentials
- Database accounts
- Administrator credentials
- Stored configuration files

11. Discovery Techniques

Collects:

- Server information
- User accounts

- Directory listings
- Network information

Commands may include:

- whoami
- dir
- ipconfig

12. Lateral Movement

Attackers may use it for:

- Accessing internal systems
- Uploading additional malware
- Moving across servers

13. Command & Control (C2)

Uses:

- HTTP/HTTPS requests
- Web-based command execution
- Remote attacker-controlled panels

14. Exfiltration

Can steal:

- Website databases
- Sensitive documents
- Credentials
- Configuration files

15. Impact Analysis

Effects:

- Full web server compromise
- Data theft
- Website defacement
- Persistent attacker access
- Internal network compromise

16. Indicators of Compromise (IOCs)

- Unknown ASP/ASPX files
- Suspicious web requests
- Unauthorized file uploads
- Unusual server activity
- Modified web directories

17. Detection & Mitigation

Detection

- Web server log monitoring
- File integrity monitoring
- Web shell detection tools
- EDR monitoring

Mitigation

- Secure file upload functionality
- Patch web applications
- Restrict web server permissions
- Use Web Application Firewall (WAF)
- Monitor server changes

18. References

- MITRE ATT&CK
- Microsoft Security Documentation
- Web shell analysis reports
- Security research articles

S0228-NanHaiShu

1. Name & Type

- **Name:** NanHaiShu
- **Type:** Remote Access Trojan (RAT) / Backdoor
- **MITRE ID:** S0228

2. Description

NanHaiShu is a Remote Access Trojan (RAT) used for cyber espionage and long-term access to infected systems. It allows attackers to execute commands, steal files, monitor victims, and communicate with remote command servers.

3. Threat Actor Association

- Associated with Chinese cyber espionage operations
- Used in targeted attacks against governments and organizations

4. Supported Platforms

- Microsoft Windows

5. ATT&CK Techniques Used

Technique	Purpose
Command Execution (T1059)	Execute remote commands
Registry Run Keys (T1547)	Persistence
Data Exfiltration (T1041)	Steal information
Remote Services (T1021)	Remote access
Obfuscated Files (T1027)	Hide malware

6. Execution Method

Delivered through:

- Spear-phishing emails
- Malicious attachments
- Exploited vulnerabilities
- Fake documents

7. Persistence Techniques

- Registry modifications
- Startup entries
- Scheduled tasks
- Service installation

8. Privilege Escalation

Attackers may:

- Exploit Windows vulnerabilities
- Use stolen administrator credentials
- Abuse weak permissions

9. Defense Evasion

- Obfuscated code
- Encrypted communications
- Hidden files and processes
- Anti-analysis techniques

10. Credential Access

Targets:

- User credentials
- Administrative accounts
- Cached passwords
- Authentication information

11. Discovery Techniques

Collects:

- System information
- Running processes
- Network configuration
- User account details

Commands used:

- whoami

- ipconfig
- tasklist

12. Lateral Movement

Uses:

- Remote Desktop Protocol (RDP)
- SMB shares
- Stolen credentials
- Remote execution tools

13. Command & Control (C2)

Communicates with attacker servers through:

- HTTP/HTTPS traffic
- Encrypted channels
- Remote shell access

14. Exfiltration

Can steal:

- Sensitive documents
- Credentials
- System information
- Internal data

15. Impact Analysis

Effects:

- Persistent remote access
- Espionage operations
- Data theft
- Network compromise
- Security monitoring bypass

16. Indicators of Compromise (IOCs)

- Suspicious outbound connections

- Unknown startup entries
- Strange processes
- Registry modifications
- Unusual network activity

17. Detection & Mitigation

Detection

- EDR monitoring
- Network traffic analysis
- Malware sandboxing
- Registry monitoring

Mitigation

- Use MFA
- Patch Windows systems
- Restrict admin privileges
- Filter phishing emails
- Monitor outbound traffic

18. References

- MITRE ATT&CK
- Security research reports
- Malware analysis documentation
- Threat intelligence reports

S0229-Orz

1. Name & Type

- **Name:** Orz
- **Type:** Remote Access Trojan (RAT) / Backdoor
- **MITRE ID:** S0229

2. Description

Orz is a Remote Access Trojan (RAT) used by attackers to remotely control infected systems. It supports command execution, file management, information theft, and persistent access.

3. Threat Actor Association

- Associated with cyber espionage and targeted attacks
- Used by advanced threat actors against organizations and government targets

4. Supported Platforms

- Microsoft Windows

5. ATT&CK Techniques Used

Technique	Purpose
Command Execution (T1059)	Execute attacker commands
Registry Run Keys (T1547)	Persistence
Remote Services (T1021)	Remote access
Obfuscated Files (T1027)	Hide malware
Data Exfiltration (T1041)	Steal sensitive data

6. Execution Method

Delivered through:

- Spear-phishing emails
- Malicious attachments
- Exploited vulnerabilities

- Social engineering attacks

7. Persistence Techniques

- Registry startup entries
- Scheduled tasks
- Service installation
- Startup folder modifications

8. Privilege Escalation

Attackers may:

- Exploit Windows vulnerabilities
- Use stolen administrator credentials
- Abuse weak permissions

9. Defense Evasion

- Obfuscated code
- Encrypted communications
- Hidden processes
- Anti-analysis techniques

10. Credential Access

Targets:

- User credentials
- Administrative accounts
- Stored passwords
- Authentication information

11. Discovery Techniques

Collects:

- System information
- Running processes
- User account details
- Network configuration

Commands used:

- whoami
- ipconfig
- tasklist

12. Lateral Movement

Uses:

- RDP access
- SMB shares
- Stolen credentials
- Remote execution tools

13. Command & Control (C2)

Communicates using:

- HTTP/HTTPS traffic
- Encrypted communication channels
- Remote shell connections

14. Exfiltration

Can steal:

- Sensitive documents
- Credentials
- Internal data
- System information

15. Impact Analysis

Effects:

- Persistent attacker access
- Data theft
- Network compromise
- Espionage activity
- Security bypass

16. Indicators of Compromise (IOCs)

- Suspicious outbound traffic
- Unknown startup entries
- Unusual registry changes
- Strange running processes
- Communication with malicious IPs

17. Detection & Mitigation

Detection

- EDR monitoring
- Network traffic analysis
- Registry monitoring
- Malware sandbox analysis

Mitigation

- Enable MFA
- Patch Windows systems
- Restrict administrator privileges
- Filter phishing emails
- Monitor suspicious network traffic

18. References

- MITRE ATT&CK
- Threat intelligence reports
- Malware analysis research
- Cybersecurity documentation

Conclusion and Summary

This report analyzed MITRE ATT&CK Campaign C0022 (Operation Dream Job) and software S0220–S0230. The campaign used phishing, fake job offers, malware, credential theft, persistence, and remote access techniques to target organizations and steal sensitive information.

The research covered multiple malware tools such as Chaos, Umbreon, POWERSTATS, Smoke Loader, sqlmap, and Orz, along with their attack methods, persistence techniques, and impacts on systems.

The report also highlighted MITRE ATT&CK techniques, detection opportunities, and defensive recommendations including MFA, EDR solutions, patch management, phishing awareness, and network monitoring to improve cybersecurity defense. Overall, the project provided practical understanding of cyber threat intelligence, malware analysis, and modern cyber attack operations.
